

CS728 Programming Assignment 3

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1 Part 1: Classical Retrieval

1.1 Comparison of Retrieval Methods

Method	Recall@1	Recall@5
BM25	0.2312	0.4114
msmarco-MiniLM	0.3882	0.6106
UAE-large-v1	0.6502	0.8842

Table 1: Recall at different ranks for the three retrieval methods.

2 Part 2: Lost-in-the-middle

2.1 Results

- Recall@1: 0.0122
- Recall@5: 0.2486

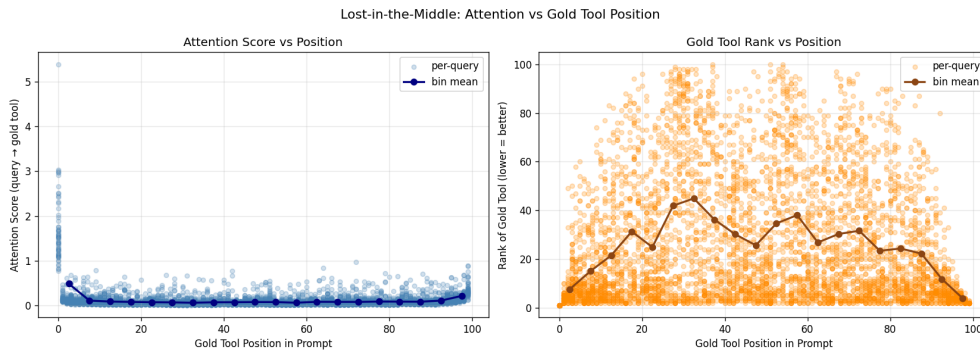


Figure 1: Position Effect on Attention Based Ranking

2.2 Plot

3 Part 3: Retrieval Heads

3.1 Head selection strategy

TO-DO

Selected Heads: [(6, 11), (7, 12), (4, 16), (5, 11), (8, 19), (10, 9), (2, 20), (10, 1), (6, 8), (10, 23), (6, 9), (5, 10), (8, 20), (8, 13), (12, 13), (5, 9), (10, 13), (10, 11), (8, 17), (10, 26)]

Note: format of selected heads is (layer number, head number).

3.2 Results

- Recall@1: 0.2632
- Recall@5: 0.5918

3.3 Comparison with previous methods